



# Mixed Reality-Assisted Motorcycle Assembly Training with Spatially Aligned Interactive Overlays

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## Abstract

This project presents the development of a Unity-based mixed reality (MR) training system designed to guide users through the process of assembling a motorcycle using real-world tools and components. By integrating depth-sensing cameras with spatial computing, the system overlays context-aware visual instructions directly onto physical objects, enabling precise alignment between virtual guidance and the user's environment. Computer vision techniques are employed to detect and track relevant tools within the user's field of view, dynamically highlighting them according to each step in the assembly workflow. In addition, 3D overlays indicate correct part placement and orientation on the physical motorcycle, reducing ambiguity and user error.



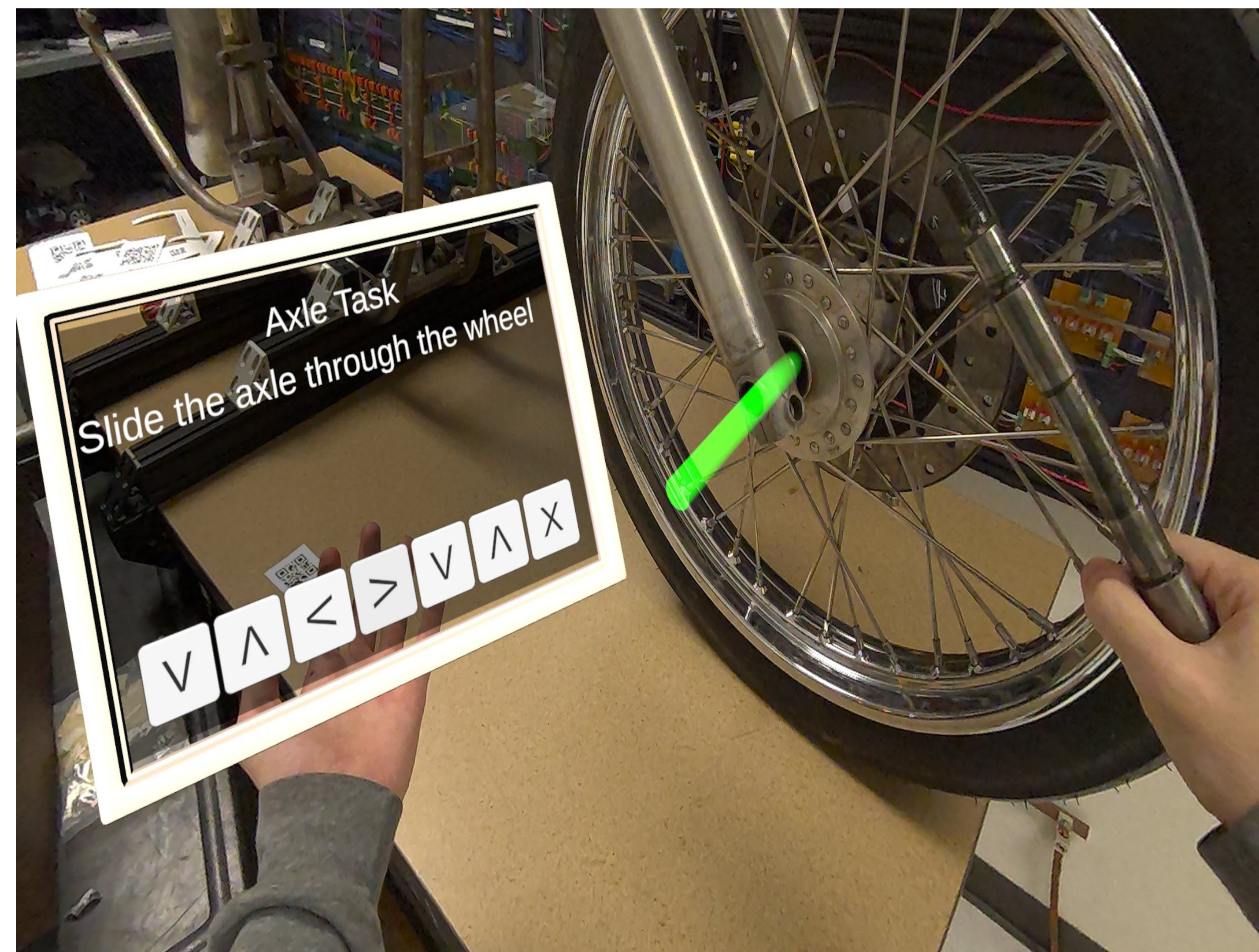
Above: A photograph of the physical motorcycle propped up for assembly.

## Methods

This project was developed using Unity and deployed on the Meta Quest 3 headset. The system leverages the Meta XR SDK and its Depth API to enable spatial awareness and alignment of virtual content with the physical environment.

A physical motorcycle serves as the assembly platform, with QR codes affixed to key components to provide stable reference points for tracking.

Users progress through a predefined series of assembly steps, with the system dynamically updating overlays to indicate required tools, part placement, and task order. Users confirm task completion using hand-tracked gestures.



Above: A first-person view of the user aligning the axle with the wheel, guided by a virtual tablet and spatial overlay indicating correct placement.

## Results

Virtual overlays remained consistently aligned with physical components upon initial viewing of QR markers, enabling accurate step-by-step guidance.

User interaction testing showed that participants were able to follow the guided assembly sequence with reduced uncertainty compared to unguided attempts.

Other step-by-step physical tasks can be virtually aided using the system used in this example.



Above: A first-person view of the user positioning the wheel onto the motorcycle, with virtual overlays properly occluded by physical objects.

## References

- A. Tang, C. Owen, F. Biocca, and W. Mou, "Comparative effectiveness of augmented reality in object assembly," in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '03), East Lansing, MI, USA, 2003, pp. 73–80, doi: 10.1145/642611.642626.
- S. J. Henderson and S. K. Feiner, "Augmented reality in the psychomotor phase of a procedural task," in 2011 10th IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Basel, Switzerland, Oct. 2011, doi: 10.1109/ISMAR.2011.6092386.
- E. A. Suma, D. M. Krum, and M. Bolas, "Sharing space in mixed and virtual reality environments using a low-cost depth sensor," in 2011 IEEE International Symposium on VR Innovation (ISVRI), Singapore, Mar. 2011, doi: 10.1109/ISVRI.2011.5759673.